

$$U_{int} = - \frac{V \hbar c}{\pi^2} \int_0^\infty \frac{k^3 e^{-\beta \hbar c k}}{1 - e^{-\beta \hbar c k}} dk$$

let $x = \beta \hbar c k$, then

$$dx = (\beta \hbar c) dk \Rightarrow dk = \left(\frac{dx}{\beta \hbar c} \right)$$

$$\begin{aligned} U_{int} &= - \frac{V \hbar c}{\pi^2} \int_0^\infty \left(\frac{x}{\beta \hbar c} \right)^3 \left(\frac{e^{-x}}{1 - e^{-x}} \right) \left(\frac{dx}{\beta \hbar c} \right) \\ &= - \frac{V \hbar c}{\pi^2 (\beta \hbar c)^4} \int_0^\infty \frac{x^3 e^{-x}}{1 - e^{-x}} dx \end{aligned}$$

* Solving this integral:

$$\text{Recall } \sum_{n=0}^{\infty} y^n = \frac{1}{1-y} \text{ for } 0 < |y| < 1$$

$$\text{then } \frac{1}{1 - e^{-x}} = \sum_{n=0}^{\infty} (e^{-x})^n = \sum_{n=0}^{\infty} e^{-x(n+1)}$$

Plugging this in:

$$\begin{aligned} \int_0^\infty x^3 e^{-x} \left(\sum_{n=0}^{\infty} e^{-x(n+1)} \right) dx &= \int_0^\infty x^3 \sum_{n=0}^{\infty} e^{-x(n+1)} dx \\ &\quad \uparrow \text{swap} \uparrow \\ &= \sum_{n=0}^{\infty} \int_0^\infty x^3 e^{-x(n+1)} dx \end{aligned}$$

let $z = x(n+1)$

$$\frac{dz}{n+1} = dx$$

$$= \sum_{n=0}^{\infty} \int_0^\infty \frac{z^3}{(n+1)^3} e^{-z} \frac{dz}{(n+1)}$$

$$= \left(\sum_{n=0}^{\infty} \frac{1}{(n+1)^4} \right) \left(\int_0^{\infty} x^3 e^{-x} dx \right)$$

$$\Gamma(4) = \int_0^{\infty} x^3 e^{-x} dx = 3!$$

(Gamma function)

$$\zeta(4) = \sum_{n=0}^{\infty} \frac{1}{(n+1)^4} = \frac{\pi^4}{90}$$

(Zeta-function)

Therefore

$$\int_0^{\infty} \frac{x^3 e^{-x}}{1 - e^{-x}} dx = \frac{\pi^4}{90} \cdot 3! = \frac{\pi^4}{15}$$